



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 AT3 Technical Presentation

Slot: A

Course code : DSA0204

Course Name: Computer Vision with OpenCV

Faculty Name: Dr. G. A. Senthil

Presented By:

Chalamala Sai Harshitha

192424346

Comparative Technical Analysis of Image Processing and Computer Vision

Introduction:

- **Image Processing (IP)** is the process of enhancing or modifying digital images.
- **Computer Vision (CV)** enables computers to understand and interpret images or videos.
- Image Processing improves image quality, whereas Computer Vision extracts meaningful information for decision-making.
- Both technologies are widely used in healthcare, surveillance, autonomous vehicles, agriculture, and robotics.

Objective:

- Compare Image Processing and Computer Vision.
- Understand their problem scope and real-world applications.

Problem Identification

Image Processing

Problem: Raw images often contain noise, blur, poor contrast, or distortion.

These issues reduce image quality.

Goal: Improve image quality for better visualization or further analysis.

Computer Vision

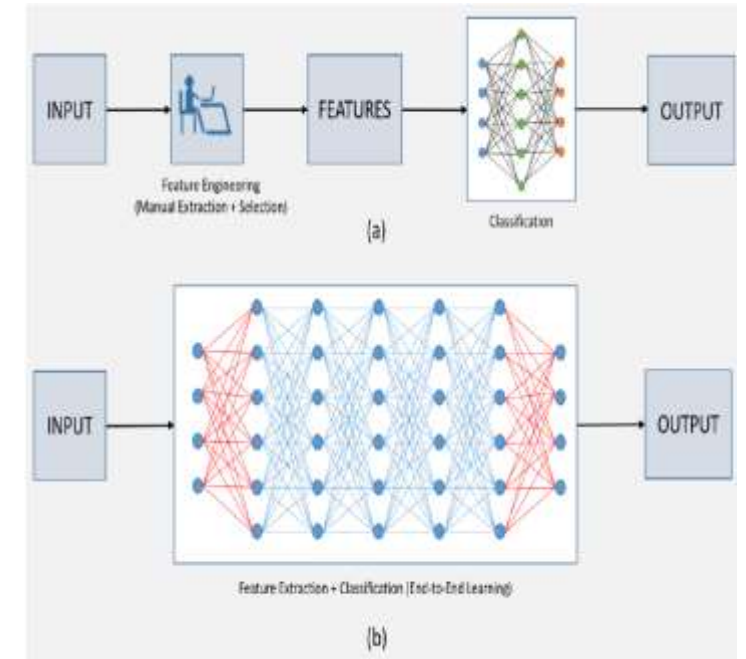
Problem: Computers cannot naturally recognize objects or understand scenes.

Images must be interpreted automatically.

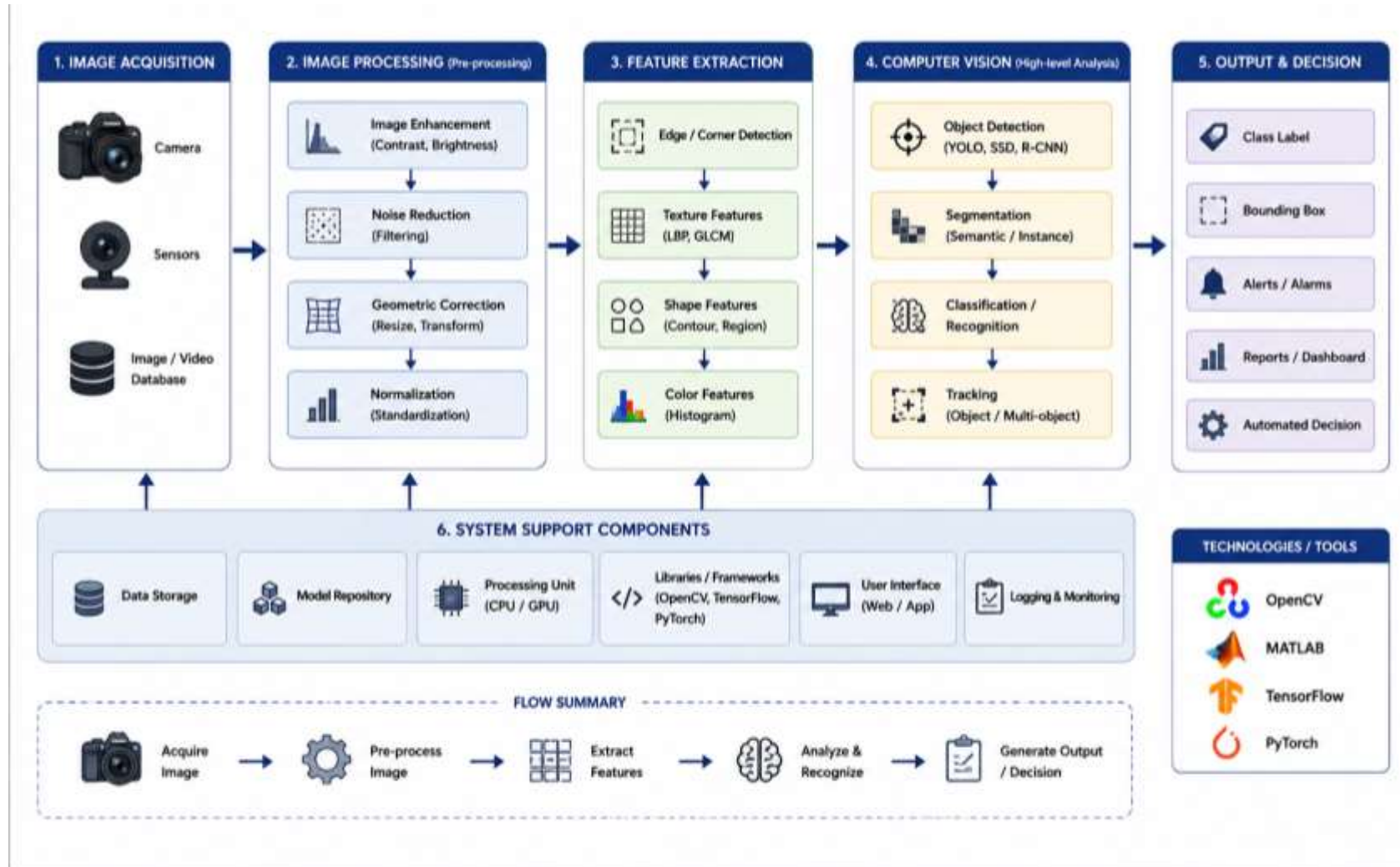
Goal: Detect, classify, recognize, and understand objects within images.

Technical Comparison

Feature	Image Processing	Computer Vision
Purpose	Enhance images	Understand images
Input	Digital image	Image or Video
Output	Enhanced image	Information/Decision
Main Tasks	Filtering, Denoising, Enhancement	Detection, Recognition, Classification
Uses	Medical image enhancement	Face recognition, Self-driving cars



Architecture Diagram



Real-World Applications

Healthcare

Enhance X-ray and MRI images

Detect diseases using AI

Surveillance

Face recognition

Intrusion detection

Autonomous Vehicles

Lane detection

Traffic sign recognition

Agriculture

Crop disease detection

Retail

Automated checkout

Customer behavior analysis

Advantages

Image Processing

- Improves image quality for better visual analysis.
- Removes noise to produce clearer images.
- Enhances visibility by adjusting contrast and brightness.

Computer Vision

- Automates decision-making using image analysis.
- Increases accuracy in object detection and recognition.
- Reduces manual effort through intelligent automation.

Conclusion

- Computer Vision has transformed industries by solving complex real-world problems.
- It improves efficiency, accuracy, and automation in healthcare, surveillance, transportation, agriculture, and retail.
- Advances in AI and Deep Learning continue to expand Computer Vision applications

Challenges

- Poor lighting conditions
- Image noise
- Occlusion of objects
- High computational requirements
- Privacy and security concerns
- Requirement of large labelled datasets

References

1. Guo, M. H., Xu, T. X., Liu, J. J., Liu, Z. N., Jiang, P. T., Mu, T. J., Zhang, S. H., Martin, R. R., Cheng, M. M., & Hu, S. M. (2022). Attention mechanisms in computer vision: A survey. *Computational Visual Media*, 8(3), 331–368.
2. Szeliski, R. (2022). *Computer vision: Algorithms and applications* (2nd ed.). Springer.
3. Raman, B., Murala, S., Chowdhury, A. S., Dhall, A., & Goyal, P. (Eds.). (2022). *Computer vision and image processing*. Springer.
4. Gupta, D., Bhurchandi, K. M., Murala, S., Raman, B., & Kumar, S. (Eds.). (2023). *Computer vision and image processing*. Springer Nature.
5. Gutti, G. K. (2024). *Neural network computer vision with OpenCV 5: Build computer vision solutions using Python and DNN module*. BPB Publications.

Computer Vision Solutions for Real-World Problems Across Multiple Domains

Introduction:

- Computer Vision (CV) is a branch of Artificial Intelligence that enables computers to capture, process, and interpret images or videos.
- It mimics human vision to identify objects, people, and activities.
- Computer Vision is widely used to automate tasks, improve accuracy, and support decision-making.

Objectives:

- Identify real-world problems in different sectors.
- Explain how Computer Vision provides effective solutions.
- Understand the importance of Computer Vision in modern applications.

Problem Identification

Real-World Challenges

Domain	Problem
Healthcare	Delayed disease diagnosis
Surveillance	Manual monitoring of CCTV footage
Autonomous Vehicles	Detecting obstacles and traffic signs
Agriculture	Identifying crop diseases
Retail	Long checkout queues and inventory management

Need for Computer Vision

- Faster analysis
- Higher accuracy
- Reduced human effort
- Real-time decision making

Computer Vision Solutions

Healthcare

Detects tumors and fractures from medical images

Assists doctors in diagnosis

Surveillance

Face recognition

Intrusion detection

Crowd monitoring

Autonomous Vehicles

Lane detection

Pedestrian detection

Traffic sign recognition

Agriculture

Crop disease detection

Fruit counting

Weed detection

Retail

Automated billing

Shelf monitoring

Customer behavior analysis

Architecture Diagram



Case Studies

Healthcare

- AI detects pneumonia from chest X-rays with high accuracy.

Surveillance

- Airports use face recognition for passenger identification.

Autonomous Vehicles

- Self-driving cars detect roads, vehicles, and pedestrians.

Agriculture

- Drones identify crop diseases early.

Retail

- Amazon Go stores use Computer Vision for cashier-less shopping.

Benefits

- High accuracy
- Faster processing
- Real-time monitoring
- Reduces manual work
- Improves productivity
- Supports intelligent decision-making

Conclusion

- Computer Vision solves real-world problems through intelligent image analysis.
- It improves accuracy, efficiency, and automation across various industries.
- AI and deep learning enable faster and smarter decision-making.
- Challenges such as noise, lighting, and high computation still exist.
- Future advancements will make Computer Vision more reliable and widely adopted.

References

1. Szeliski, R. (2022). *Computer vision: Algorithms and applications* (2nd ed.). Springer.
2. Gupta, D., Bhurchandi, K. M., Murala, S., Raman, B., & Kumar, S. (Eds.). (2023). *Computer vision and image processing*. Springer Nature.
3. Gutti, G. K. (2024). *Neural network computer vision with OpenCV 5: Build computer vision solutions using Python and DNN module*. BPB Publications.
4. Guo, M. H., Xu, T. X., Liu, J. J., Liu, Z. N., Jiang, P. T., Mu, T. J., Zhang, S. H., Martin, R. R., Cheng, M. M., & Hu, S. M. (2022). Attention mechanisms in computer vision: A survey. *Computational Visual Media*, 8(3), 331–368.
5. Kumar, A., Mozar, S., & Tanwar, S. (Eds.). (2024). *Artificial intelligence and computer vision for smart applications*. CRC Press.

Key Factors Affecting Image Quality in Computer Vision

Introduction:

- Image quality refers to how accurately an image represents a real-world scene.
- High-quality images improve the performance of Computer Vision systems.
- Poor image quality leads to incorrect object detection and recognition.

Objectives:

- Identify factors affecting image quality.
- Analyze their impact on Computer Vision.
- Explore methods to improve image quality.

Problem Identification

Common Causes

- Poor lighting conditions
- Sensor noise
- Low image resolution
- Lens distortion
- Motion blur
- Environmental conditions (fog, rain, dust)

Effects

- Incorrect object detection
- Reduced recognition accuracy
- Poor system reliability
- Increased processing time

Key Factors Affecting Image Quality

1. Image Acquisition Conditions

Camera position

Lighting

Focus

Exposure

2. Spatial Resolution

Number of pixels in an image.

Higher resolution provides more details.

3. Noise

Random unwanted signals introduced during image capture.

Examples: Gaussian noise, Salt-and-Pepper noise.

4. Lens Distortion

Barrel distortion

Pincushion distortion

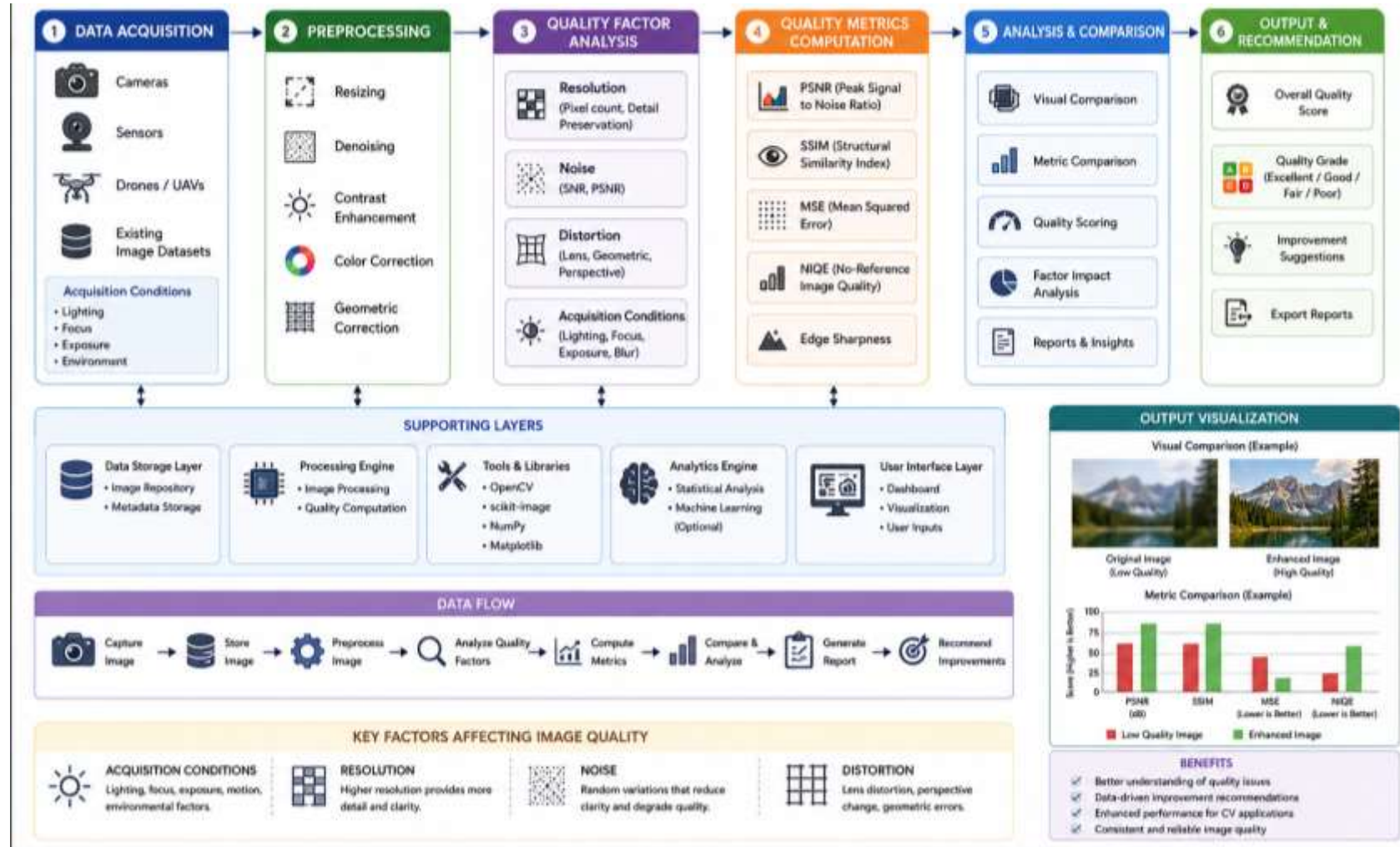
Analysis Using Visual Tools

Factor	Good Quality	Poor Quality
Resolution	High detail	Loss of details
Lighting	Proper brightness	Dark or overexposed
Noise	Clear image	Grainy image
Distortion	Straight lines	Curved lines

Analytical Tools

- Histogram
- Image comparison charts
- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index)

Architecture Diagram



Real-World Applications

Healthcare

Poor X-ray quality may lead to incorrect diagnosis.

Surveillance

Low-light CCTV images reduce face recognition accuracy.

Autonomous Vehicles

Rain and fog affect lane and obstacle detection.

Agriculture

Blurred drone images reduce crop disease detection accuracy.

Industrial Inspection

Noise can hide manufacturing defects.

Solutions and Best Practices

Hardware Improvements

High-resolution cameras

Better image sensors (CCD/CMOS)

Proper lighting systems

Software Techniques

Noise filtering

Contrast enhancement

Histogram equalization

Image sharpening

Lens distortion correction

Conclusion

- Image quality is a critical factor in Computer Vision performance.
- Poor acquisition conditions, low resolution, noise, and distortion significantly reduce system accuracy.
- Using advanced cameras, proper lighting, and image enhancement techniques improves image quality and overall Computer Vision performance.
- High-quality images lead to more reliable and efficient real-world applications.

References

1. Szeliski, R. (2022). *Computer vision: Algorithms and applications* (2nd ed.). Springer.
2. Raman, B., Murala, S., Chowdhury, A. S., Dhall, A., & Goyal, P. (Eds.). (2022). *Computer vision and image processing*. Springer.
3. Gupta, D., Bhurchandi, K. M., Murala, S., Raman, B., & Kumar, S. (Eds.). (2023). *Computer vision and image processing*. Springer Nature.
4. Guti, G. K. (2024). *Neural network computer vision with OpenCV 5: Build computer vision solutions using Python and DNN module*. BPB Publications.
5. Guo, M. H., Xu, T. X., Liu, J. J., Liu, Z. N., Jiang, P. T., Mu, T. J., Zhang, S. H., Martin, R. R., Cheng, M. M., & Hu, S. M. (2022). Attention mechanisms in computer vision: A survey. *Computational Visual Media*, 8(3), 331–368.

Thank You